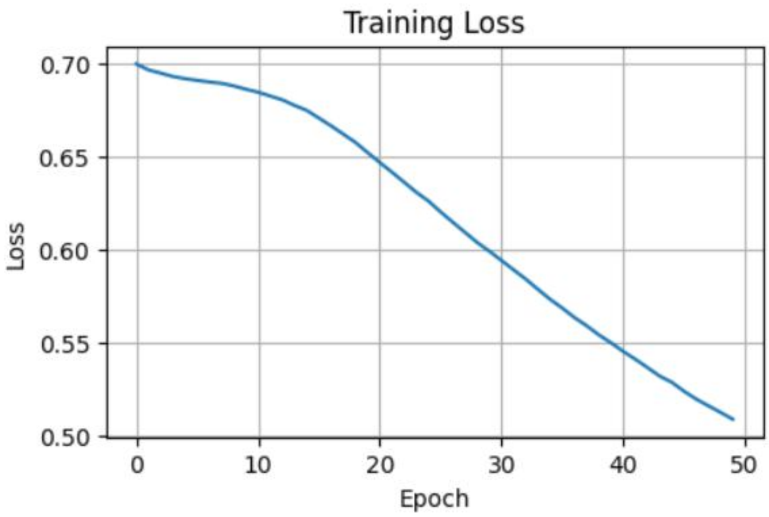
25^{th} Experiment

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import make_circles
from sklearn.metrics import accuracy_score
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

# Generate Circular Dataset
X, y = make_circles(
    n_samples=500,
    noise=0.1,
    factor=0.4,
    random_state=42
)

# Build Neural Network
model = Sequential([
    Dense(16, activation='tanh', input_shape=(2,)),
    Dense(8, activation='tanh'),
    Dense(1, activation='sigmoid')
])

# Compile Model
model.compile(
    optimizer='adam',
    loss='binary_crossentropy',
    metrics=['accuracy']
)
```

```python
# Train Model
history = model.fit(
    X, y,
    epochs=50,
    verbose=0
)

# Training Loss Graph
plt.figure(figsize=(5,3))
plt.plot(history.history['loss'])
plt.title("Training Loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.grid(True)
plt.show()

# Prediction
pred = model.predict(X)
pred_class = (pred > 0.5).astype(int)

# Accuracy
acc = accuracy_score(y, pred_class)
print("Accuracy:", round(acc*100,2), "%")

# Circular Dataset Plot
plt.figure(figsize=(5,4))
plt.scatter(X[:,0], X[:,1], c=y)
plt.title("Circular Data (Tanh Activation)")
plt.xlabel("Feature 1")
```

```
plt.ylabel("Feature 2")

plt.show()
```




```
16/16 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step
Accuracy: 81.6 %
```